

Arya Hosseini

Montreal, QC, Canada

[Portfolio](#), [GitHub](#)

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SUMMARY

Junior Game Developer with hands on experience in **Unity and Unreal Engine**. Skilled in gameplay programming, AI systems, UI implementation, and data driven design, with a foundation in C++, C#, and scripting languages. Bringing strong problem solving skills and a focus on writing maintainable, scalable code. Seeking opportunities to tackle technical challenges and contribute to building high quality, interactive games.

SKILLS

Engines & Tools: Unity, Unreal Engine, SFML, ImGui, Git

Programming Languages: C#, C++ , Lua, SQL, JavaScript, HTML/CSS

Languages: English, French, Kurdish

PROFESSIONAL EXPERIENCE

Unity Gameplay Programmer Intern - Baobab Games

May 2024 - Aug 2024

DION – Multiplayer Survival

Montreal.Canada

- Expanded the base-building system by extending existing tools and plugins. Designed a modular interaction system using interfaces, allowing interaction with world objects; integrated 18 new building parts with unique placement constraints, enabling rapid integration and testing of new building content without engineering involvement.
- Developed a network-synced item storage system based on ScriptableObjects; integrated it with base-building, world generation, UI, and inventory systems. Optimized real-time item spawning with pooled 3D physics objects for smooth, synchronized client-server interactions.
- Collaborated with artists to integrate 3D assets and streamline implementation workflows; worked with QA to address feature feedback and participated in Agile sprints to improve iteration speed and cross-team coordination.

PERSONAL PROJECT

Hallowfall - Indie Project

2024 - Present

Action Roguelike - Unity (Solo Project)

[View Project](#)

- Designed a scalable **state-based player architecture** using the State Pattern, with modular systems for movement, input, combat, and animation, ensuring extensibility and clean separation of concerns.
- Built a **data-driven ability system** with ScriptableObjects and interface-driven polymorphism, supporting randomized upgrades, composable abilities, and seamless UI integration.
- Developed a **procedural world generation system** using zone-based architecture, binary space partitioning, and multi-layer tilemap rendering; optimized performance with zone culling and coroutine-based batching.
- Implemented a **custom Flow Field Pathfinding system**, replacing Unity NavMesh to support dynamic procedural zones; enabled large-scale navigation with cost fields, flow vectors, and efficient multi-agent queries.
- Created an **adaptive Enemy AI system** mirroring the player architecture, featuring FSM-driven states, event-based communication, configurable behaviors via ScriptableObjects, and strategy-pattern-based state variations. Designed the system for **scalability**, supporting both simple enemies and complex multi-phase bosses by composing reusable states and strategies.

DEC – Computer Science, Video Game Programming

2022 – 2024

LaSalle College

Montreal.Canada

Bachelor – Engineering Science

2021 – 2022

University of Tehran

Tehran, Iran